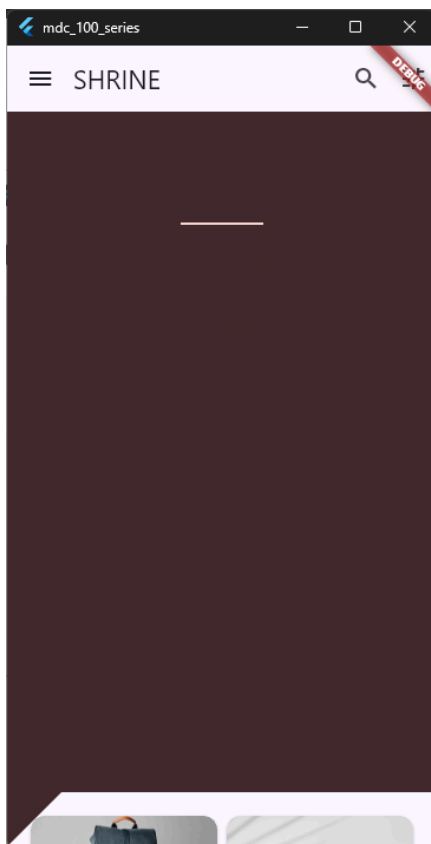
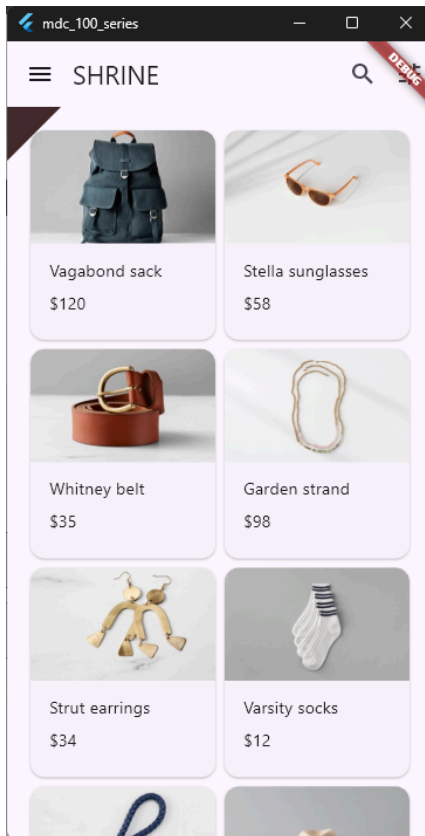
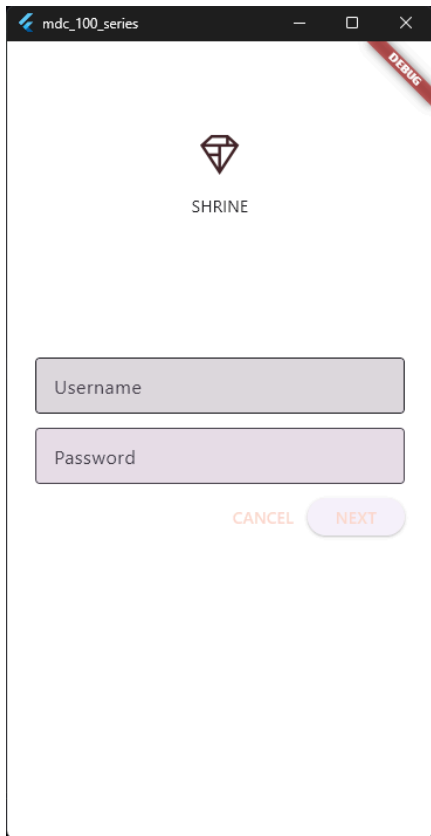




Design Systems: Using Flutter's ThemeData acts as a single source of truth for the app's entire UI. Instead of hardcoding colors, fonts, or button styles into dozens of individual widgets, ThemeData injects these properties globally down the widget tree. If a client or designer wants to change the primary brand color, we only have to update it in one place (the MaterialApp theme property) rather than manually tracking down and replacing hex codes across multiple files, ensuring consistency and adhering to the DRY (Don't Repeat Yourself) principle.

State & Animation: In MDC-104, the Backdrop relies on a StatefulWidget and an AnimationController to manage the interaction between the back layer (menu) and the front layer (products). The state holds the currently selected category. When a user taps a category in the back menu, a callback updates the state, triggering a rebuild. The front layer listens to this state change to filter the displayed products, while the AnimationController seamlessly translates the front layer's position up or down to reveal or hide the menu.